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PROJECT REPORT

ON

**“RETINOPATHY DETECTION USING MACHINE
LEARNING”**

Submitted In Partial Fulfillment of The Requirements
For

**VI SEMESTER
BACHELOR OF COMPUTER APPLICATIONS**

SUBMITTED BY

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CERTIFICATE

This is to certify that the project entitled **“Retinopathy Detection Using Machine Learning”** has been successfully submitted by **RITHIKA** RegNo. **U23IA22S0022**.

A Bonafide student at Government Degree College, Yadgir in partial fulfillment of award of Bachelor of Computer Applications in the year 2024 – 2025 is record of student own work carried out under my/our guidance. It is certified that all corrections/suggestion indicated for internal assessment have been incorporated in the report & one copy of it being deposited in the Government Post-Graduation Library. The project report has been approved as satisfies the academic requirements in respect of project work prescribed for the said bachelor's degree. It is further understood that this certified the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion draw there in but approved the project only for the purpose for which it is submitted.

Examiners:-

1.

2.

Guide
Name & Guide signature

HOD
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CANDIDATE DECLARATION

RITHIKA a student of Bachelor's in Computer Applications department bearing Regno **U23IA22S0022** of **Government Degree College, Yadgir** hereby declare that my full own responsibility for the information, results and conclusions provided in this work titled **“Retinopathy Detection Using Machine Learning”** submitted to **Adikavi Maharishi Valmiki University, Raichur** for the award of Bachelor's Degree In Computer Applications.

To the best of my knowledge, this project has not been submitted in part or full elsewhere in any other institution/organization for the award of any certificate/diploma/degree. I have completely taken care of acknowledging the contribution of others in academic work. I further declare that in any case the candidate will be slowly responsible for the same.

Date:-

Place:-

Signature of the candidate

Name:- RITHIKA

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Abstract

Retinopathy detection is a crucial task in medical imaging, aiming to identify retinal abnormalities that can lead to vision impairment or blindness. This project develops an automated deep learning-based system to detect and classify retinopathy from retinal fundus images, assisting ophthalmologists in early diagnosis and treatment planning.

The model follows a **Convolutional Neural Network (CNN)**-based approach, utilizing a pre-trained architecture such as **ResNet-50, InceptionV3, or EfficientNet** to extract high-level image features. These extracted features are then processed through fully connected layers to classify images into different severity levels of retinopathy (No Retinopathy, Mild, Moderate, Severe, and Proliferative Retinopathy). To enhance performance, techniques like **data augmentation, contrast enhancement, and image normalization** are applied during preprocessing.

To handle the challenge of imbalanced datasets, **class weighting and synthetic data generation** (via GANs or SMOTE) are employed. The model is trained on large publicly available retinal image datasets, such as **APTOS, EyePACS, or the Messidor dataset**, ensuring a diverse and robust training set. The classification performance is evaluated using standard metrics, including **accuracy, precision, recall, F1-score, and AUC-ROC**, providing a comprehensive assessment of the system's effectiveness.

The proposed system has significant real-world applications, including **early diabetic retinopathy detection, telemedicine integration, and AI-assisted ophthalmology tools** for screening large populations. The results demonstrate that deep learning-based automated diagnosis can improve **early detection rates and patient outcomes**, reducing the burden on medical professionals. This project highlights the potential of AI-driven solutions in **enhancing accessibility to healthcare and improving diagnostic accuracy** in ophthalmology.

Keywords: -

Retinopathy, Deep Learning, Convolutional Neural Networks, Fundus Image Analysis, Automated Diagnosis, Medical Imaging, Ophthalmology, Diabetic Retinopathy, AI in Healthcare, Image Classification, Early Detection, Computer-Aided Diagnosis.

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